Trefoil Divisibility: When is a Knot Divisible by the Trefoil?

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Essential Prerequisites in Knot Theory

Definition

A knot can be thought of as a rope or a string that is tangled, then tied on the ends. The simplest knot is an unknotted loop (a circle), which is called the unknot.

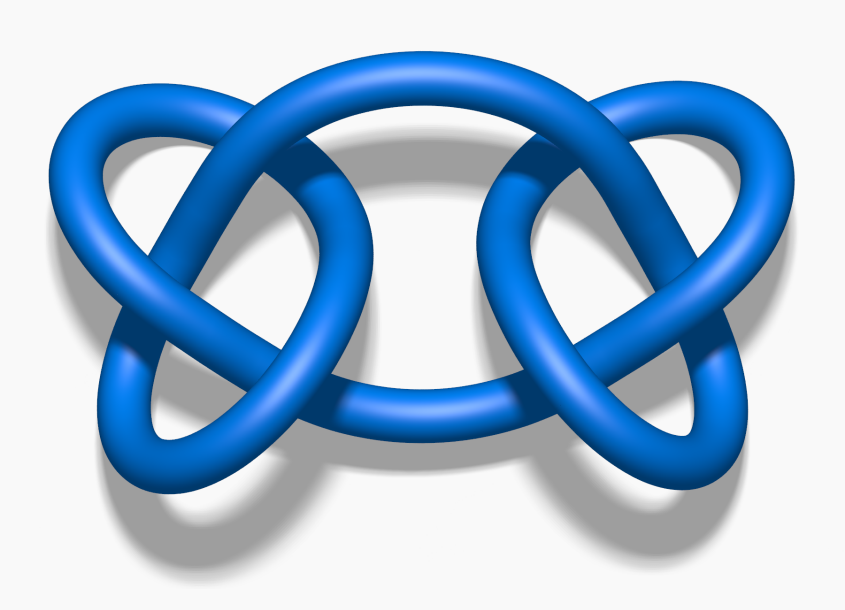


Figure: Square Knot

Definition of Prime Knot

A prime knot is unique in that it is not composed of a combination of any other existing knots, similar to prime numbers

Definition of a Composite Knot

A composite knot is composed of existing prime knots, just as composite numbers are. This composition is the knots being literally connected as shown in the image.

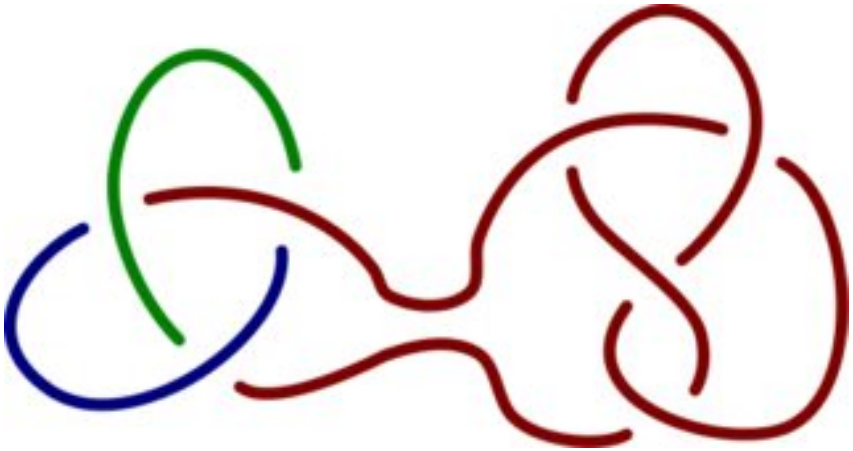


Figure: Composite Knot

Definition of a Trefoil Knot

The trefoil knot is the most basic form of prime knot, consisting of exactly three crossings.

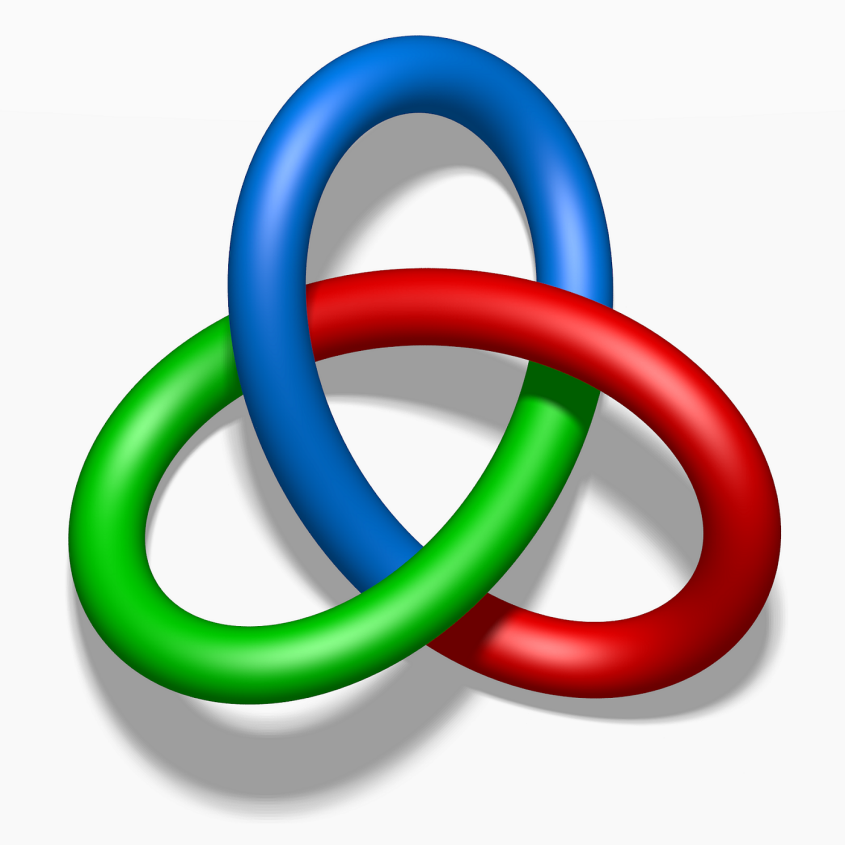


Figure: Trefoil Knot

Definition of a Knot Crossing

A knot crossing is simply the intersection point that occurs throughout the knot, which contains both positive and negative directions. The trefoil knot demonstrated previously contains exactly three crossings.

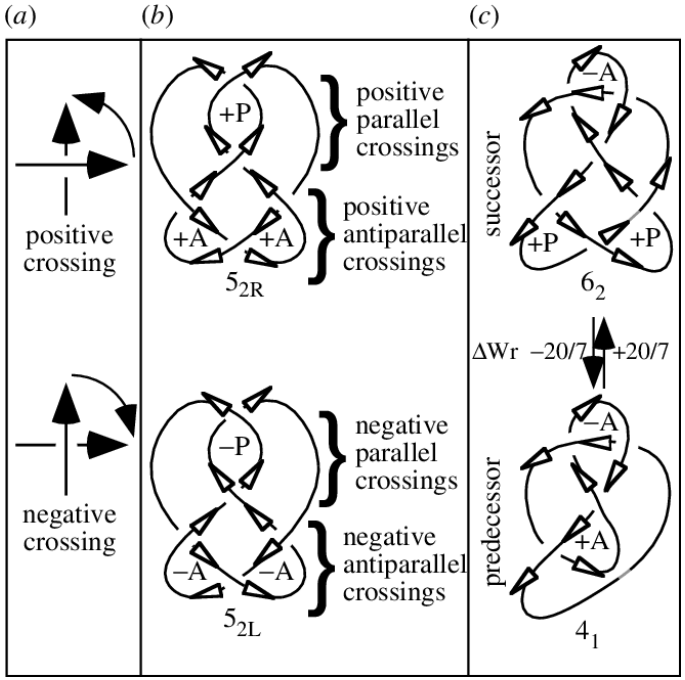


Figure: Positive and Negative Knot Crossings

Research Question

What are the criteria for trefoil knot divisibility?

This presentation presents our attempt at developing a criterion for determining when a knot is divisible by the trefoil, that is, when it can be decomposed into a connected sum involving the trefoil as a factor.

Tricolorability

University of Chicago Definition

As the word sounds, it's when 3 colours can be drawn to represent the knot, as shown here with the trefoil knot. A projection of a knot K is tricolorable if each of the strands in the projection can be colored in one of three different colors such that at each crossing either all three colors come together or only one does. This may seem trivial, but it is important to recognize these properties, and how they can contribute to divisibility.

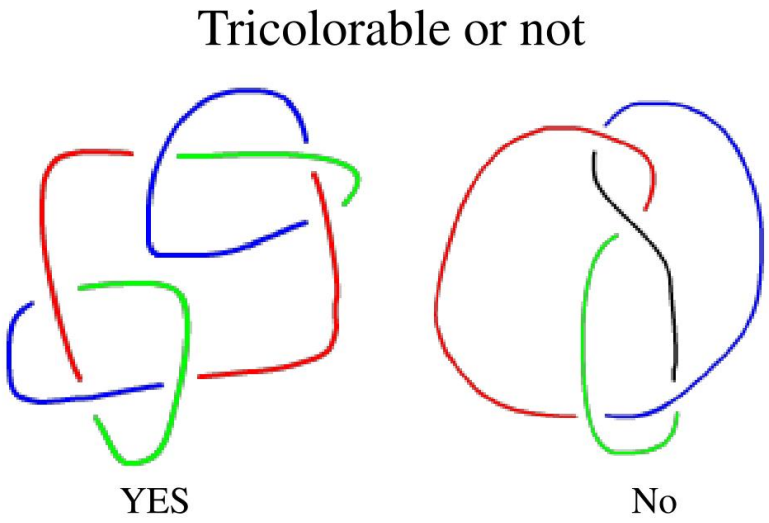


Figure: Tricolorablility diagram

Declaration

All knots that are divisible by the trefoil MUST also be tricolorable, as the trefoil knot is tricolorable. This does NOT mean that all tricolorable knots are divisible by the trefoil.

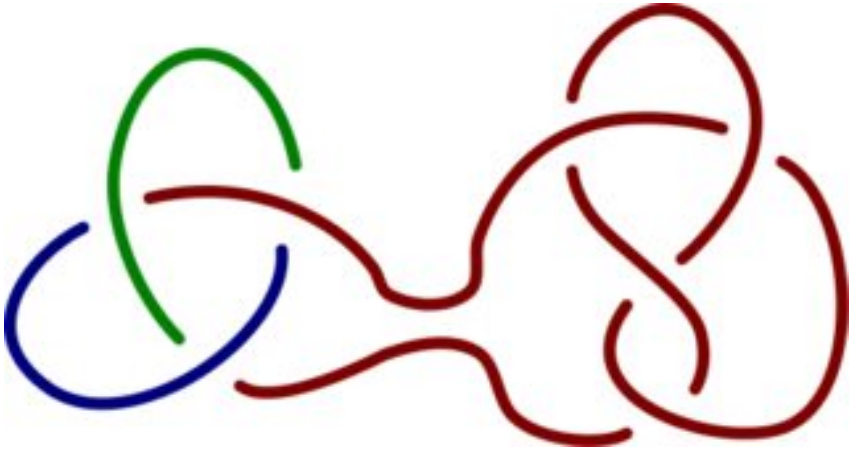


Figure: Knot divisible by trefoil

If a knot is NOT tricolorable, we know that it can't be divisible by the trefoil.

Polynomial Invariants of Knots

Conway vs. Alexander Polynomials

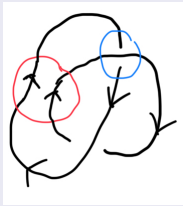
An Alexander polynomial of a knot K is normalized if $AK(t) = AK(t-1)$ and $A(1) = 1$. A normalized Alexander polynomial can be written in terms of $z = t^{1/2} - t^{-1/2}$ where only positive powers of z appear. We then denote this representation as $\nabla K(z)$ and it is called the Conway polynomial (or the potential function) of K. This is also sometimes called the Alexander-Conway polynomial.

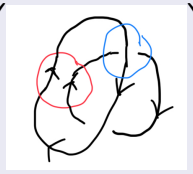
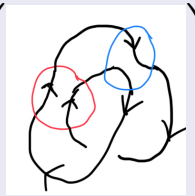
Skein Relation

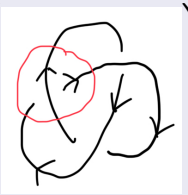
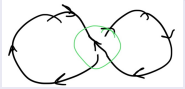
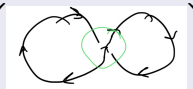
$z\nabla$ $= \nabla$ $- \nabla$

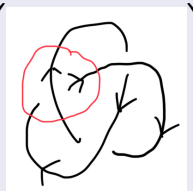
Proof of the Trefoil Conway

The Conway polynomial can be calculated using the equation

$\nabla\Big($ $\Big) = \nabla\Big($ $\Big) + z\nabla\Big($  $\Big)$

$\nabla\Big($ $\Big) = 1 + z\Big(\nabla\Big($  $\Big) + z\nabla\Big($  $\Big)\Big)$

$\nabla\Big($  $\Big) = 1 + z\Big(\nabla\Big($  $\Big) - \nabla\Big($  $\Big) + z\Big)$

$\nabla\Big($  $\Big) = 1 + z^2$

Proposed criteria for composite knot divisibility by the trefoil

If K and J are knots, then the Conway polynomial of the sum of K and J satisfies

$$\nabla_{K\#J} = \nabla_K \nabla_J$$

Therefore, if a composite knot "contains" a knot K, then its polynomial is divisible by ∇_K.

Unfortunately the converse implication is not true in general

Counterexample: Nonafoil prime knot / trefoil prime knot

$$x^8 + 7x^6 + 15x^4 + 10x^2 + 1 = (x^2 + 1)(x^6 + 6x^4 + 9x^2 + 1)$$

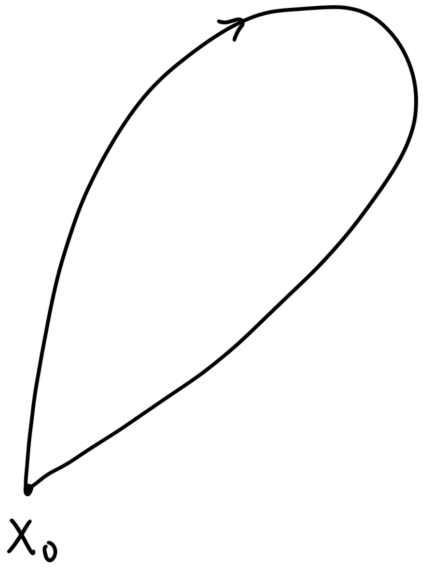
*The nonafoil knot is prime and cannot "contain" the trefoil despite it doing so algebraically

Applications of Group Theory

The Fundamental Group

Definition

A loop in a space X is a path $\alpha : [0,1] \to X$ such that $\alpha(0) = \alpha(1)$. The element $\alpha(0) = \alpha(1)$ of X_o is the base point of loop α.

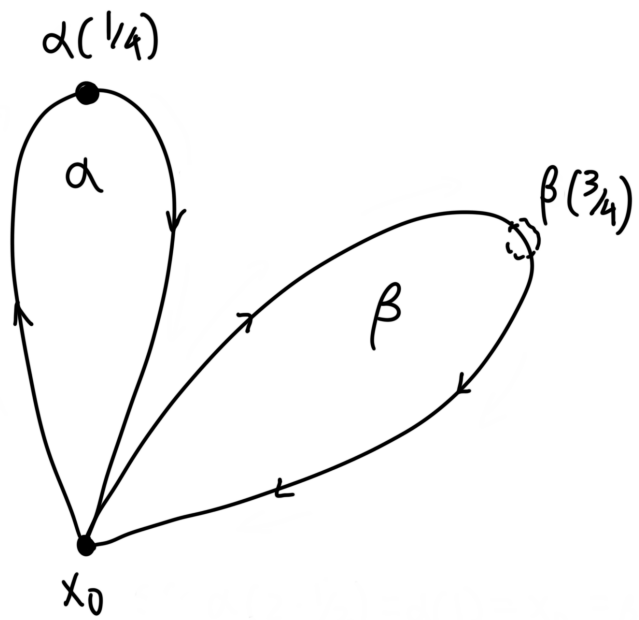


The Fundamental Group

Definition

Suppose the loops α and β in a space X have a common base point x_0. The concatenation of α and β is the path denoted $\alpha * \beta$ and defined by

$$(\alpha * \beta) = \begin{cases} \alpha(2s) & \text{for } 0 \leq s \leq \frac{1}{2} \\ \beta(2s-1) & \text{for } \frac{1}{2} \leq s \leq 1 \end{cases}$$



The Fundamental Group

Definition

Suppose the loops α and β in a space X have a common base point x_0. A homotopy from α to β is a continuous function $H : [0,1] \times [0,1] \to X$ such that the loops $H_t : [0,1] \to X$ defined by $H_t(s) = H(s,t)$ all have base point x_0 and such that $H_0 = \alpha$ and $H_1 = \beta$. We say a loop α is homotopic to the loop β if and only if there is such a homotopy from α to β. This relation is denoted $\alpha \sim \beta$.

Figure: A loop that is being deformed.

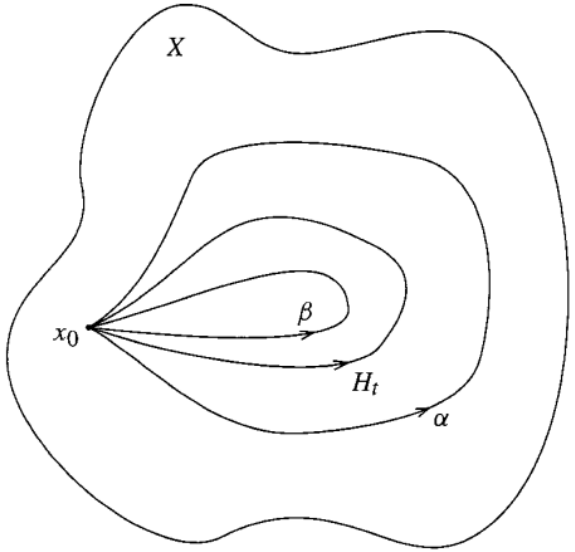


Figure: Homotopy from α to β.

The Fundamental Group

Definition

For a loop α with base point x_0 in a space X, the set of all loops homotopic to α is the homotopy class of α. This set is denoted $\langle\alpha\rangle$

These loop relations create the Fundamental Group of Knots.

Definition

Let X be a topological space with base point x_0. The set of homotopy classes of loops in X based at x_0 along with the operations of forming products of homotopy classes is the fundamental group of the space based at x_0. This group is denoted $\pi_1(X, x_0)$.

Definition

A knot is an embedding of a circle into 3-dimensional Euclidean space. The knot group of a knot K is defined as the fundamental group of the knot complement of K in $\mathbb{R}^3$, $\pi_1(\mathbb{R}^3 \backslash K)$

The Fundamental Group

Properties

The Fundamental Group satisfies the definition of a group:

1. Associativity: for any loop α, β and γ in X, $(\langle\alpha\beta\rangle)\langle\gamma\rangle = \langle\alpha\rangle(\langle\beta\gamma\rangle)$
2. Identity Element: there exists a constant path ϵ where any value of $\epsilon(s)$ is at x_0, satisfying $\langle\alpha\rangle\langle\epsilon\rangle = \langle\alpha\rangle = \langle\epsilon\rangle\langle\alpha\rangle$
3. Inverses: For any loop α, the reverse loop α^{-1} defined by $\alpha^{-1}(s) = \alpha(1-s)$ satisfies $\langle\alpha\rangle\langle\alpha^{-1}\rangle = \langle\epsilon\rangle = \langle\alpha^{-1}\rangle\langle\alpha\rangle$

Surjective Homomorphism

Definition

Surjection: Surjective if every element of Y can be mapped to at least one element of X.

Group Homomorphisms: A way to map two algebraic structures such that its structure is preserved. Let there be two operations $\triangle$ and $\blacksquare$ of the groups A and B respectively. Then, a group homomorphism f: A $\rightarrow$ B is a function f such that $f(x \triangle y) = f(x) \blacksquare f(y)$ for all $x, y \in A$.

Important Theorems

Crowell-Fox Theorem A

If there exists a surjective homomorphism from $G(K_1)$ to $G(K_2)$, then the Alexander Polynomial of K_1 is divisible by that of K_2.

Kitano-Suzuki Theorem

Let K be a prime knot and G(K) its knot group. It is well known that a partial order can be defined on the set of prime knots as follows: for two knots K_1, K_2, we write $K_1 \geq K_2$ if there exists a surjective group homomorphism from $G(K_1)$ onto $G(K_2)$.

Theorem 1.1. *The partial order "$\geq$" on the knots in Rolfsen's table is given by*

$$\left.\begin{array}{l} 8_5, 8_{10}, 8_{15}, 8_{18}, 8_{19}, 8_{20}, 8_{21}, 9_1, 9_6, 9_{16}, 9_{23}, \\ 9_{24}, 9_{28}, 9_{40}, 10_5, 10_9, 10_{32}, 10_{40}, 10_{61}, 10_{62}, \\ 10_{63}, 10_{64}, 10_{65}, 10_{66}, 10_{76}, 10_{77}, 10_{78}, 10_{82}, \\ 10_{84}, 10_{85}, 10_{87}, 10_{98}, 10_{99}\ 10_{103}, 10_{106}, 10_{112}, \\ 10_{114}, 10_{139}, 10_{140}, 10_{141}, 10_{142}, 10_{143}, 10_{144}, \\ 10_{159}, 10_{164} \end{array}\right\} \geq 3_1,$$

$$\left.\begin{array}{l} 8_{18}, 9_{37}, 9_{40}, 10_{58}, 10_{59}, 10_{60}, 10_{122}, 10_{136}, \\ 10_{137}, 10_{138} \end{array}\right\} \geq 4_1,$$

$$10_{74}, 10_{120}, 10_{122} \geq 5_2.$$

Important Theorems

Crowell-Fox Chapter VI, Exercise 6

Let us say that a knot diagram has property *I* if it is possible to color the projected overpasses in three colors, assigning a color to each edge in such a way that

1. the three overpasses that meet at a crossing are either all colored the same or are all colored differently
2. all three colors are actually used

Show that a diagram of a knot K has property *I* if and only if the group of K can be mapped homomorphically onto the symmetric group of degree three.

Crowell-Fox Theorem B

Crowell-Fox shows that tricolorability implies the group of K has a surjective homomorphism on S_3.

Torus Knots & Polynomials (deviation from trefoil divisibility in the classical sense)

What are Torus Knots?

A torus knot is a special kind of knot that lies on the surface of an unknotted torus in R^3.

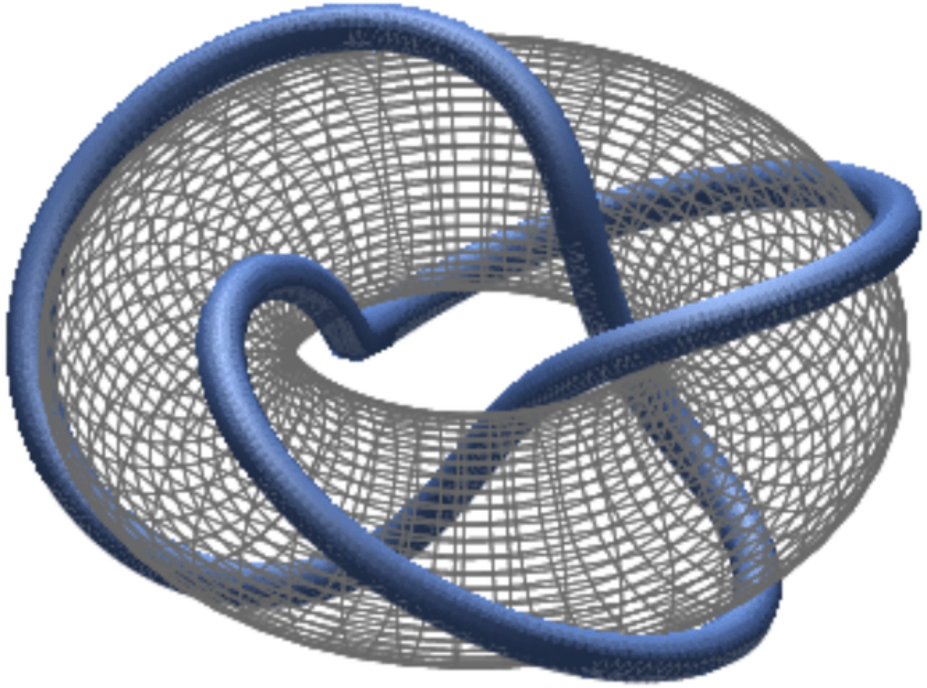


Figure: Trefoil torus knot lying on the surface of a torus

Torus Knot Conway Polynomials

$T(3,2) \rightarrow x^2 + 1$

$T(5,2) \rightarrow x^4 + 3x^2 + 1$

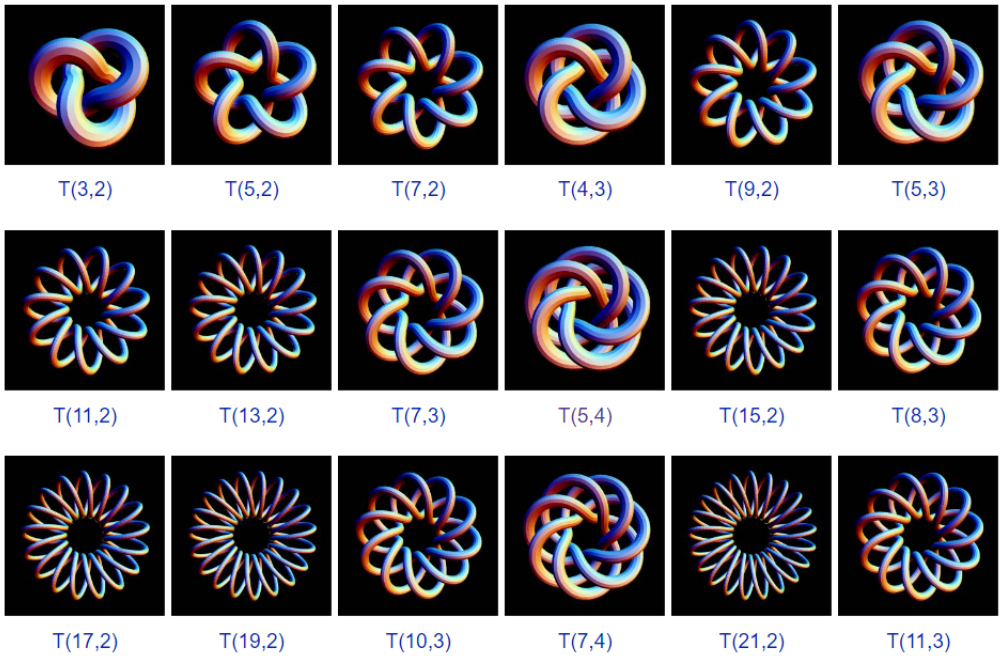
$T(7,2) \rightarrow x^6 + 5x^4 + 6x^2 + 1$

$T(4,3) \rightarrow x^6 + 5x^4 + 5x^2 + 1$

$T(9,2) \rightarrow x^8 + 7x^6 + 15x^4 + 10x^2 + 1$

$T(5,3) \rightarrow x^8 + 7x^6 + 14x^4 + 8x^2 + 1$

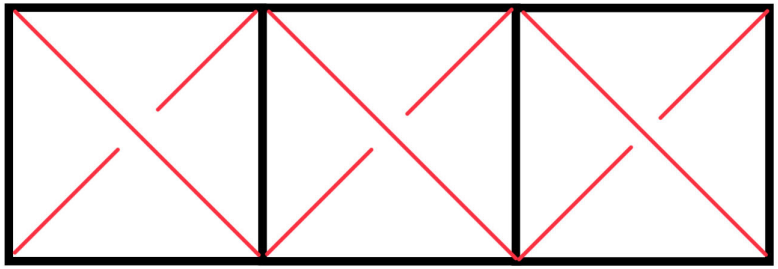
Figure: Table of torus knots



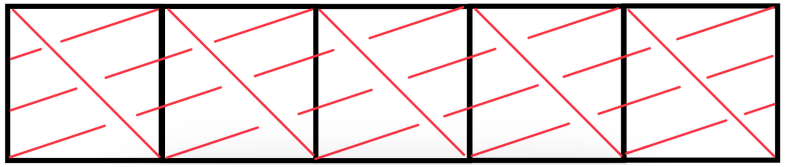
Notation

A torus knot is denoted by its braid representation, T(p,q) where p is the braid length and q is the number of strands

e.g.



(a) T(3,2)



(b) T(5,4)

In order to obtain the knot "wrap" into a circle/torus in R^3 joining the ends.

Note

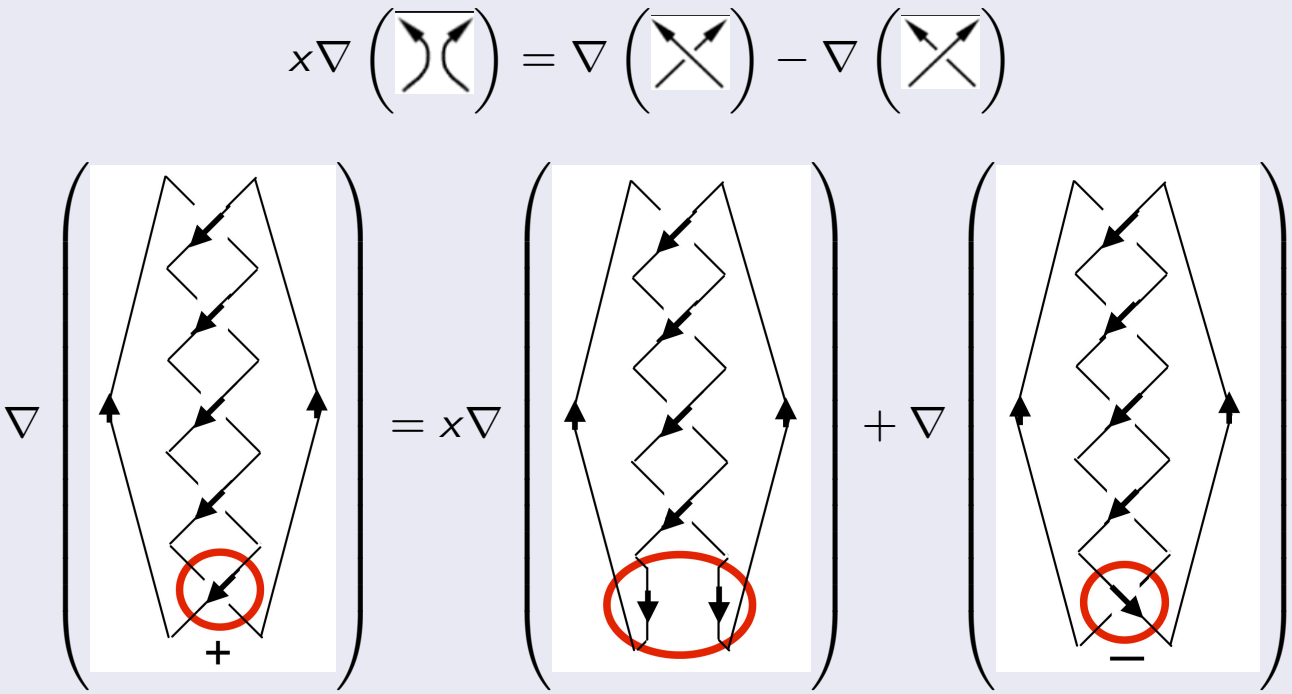
p & q cannot be coprime or it is a torus *link*.

Theorem

Torus knots T(p,2), when $p(\mathrm{mod}\ 3) \equiv 0$ its a prime knot whose Conway polynomial is divisible by the trefoil polynomial $x^2 + 1$

Proof Sketch

Using the Skein relation and torus knots T(5,2) and T(3,2),



Proof Sketch

From the braid manipulations, we get the expression,

$$\nabla_p = x\nabla_{p-1} + \nabla_{p-2}$$

For divisibility by the trefoil $(x^2 + 1)$,

$$x^2 + 1 = 0$$

$$x = \sqrt{-1}$$

$$x = i$$

Therefore substituting i into x will determine the remainder through polynomial division,

Building a sequence,

$$\underline{\mathbf{0}, i}, -1, \mathbf{0}, -1, -i, \mathbf{0}, -i, 1, \mathbf{0}, 1, i, \underline{\mathbf{0}, i}, \ldots$$

□

Theorem

Torus knots T(p,3), where $p \rightarrow$ even its a prime knot whose Conway polynomial is divisible by the trefoil polynomial $x^2 + 1$

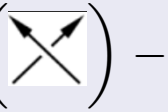
Proof Sketch

From the Livingston torus knot Alexander polynomial formula,

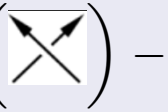
$$\Delta_{T(p,q)}(t) = \frac{(t^{|pq|} - 1)(t - 1)}{(t^{|p|} - 1)(t^{|q|} - 1)}$$

Converting from Alexander invariant to Conway invariant, using their respective Skein relation definitions,

For the Alexander,

$(t^{1/2} - t^{-1/2})\nabla$ $= \nabla$  $- \nabla$

For the Conway,

$x\nabla$ $= \nabla$  $- \nabla$

Proof Sketch

Therefore,

$$x = t^{1/2} - t^{-1/2}$$

$$t = \frac{x^2 \pm x\sqrt{x^2+4}}{2} + 1$$

Substituting into the Livingston equation evaluating x=i & evaluating for $q = 3$ from T(p,3),

Expression equal to zero when p is even proved by induction $\square$

The Torus Knots Whose Conway Polynomial is Divisible by the Trefoil Polynomial $x^2 + 1$

Because the trefoil $x = i$ is a root of unity and its Conway polynomial is cyclic every 6 cycles, it can be summarized,

- None of $\nabla_{T(p,6q+1)}$ are divisible
- $\nabla_{T(p(\mathrm{mod}\ 3)\equiv 0,6q+2)}$ are divisible e.g. $\nabla_{T(39,14)}$
- $\nabla_{T(p(\mathrm{mod}\ 2)\equiv 0,6q+3)}$ are divisible e.g. $\nabla_{T(44,39)}$
- $\nabla_{T(p(\mathrm{mod}\ 3)\equiv 0,6q+4)}$ are divisible e.g. $\nabla_{T(57,40)}$
- None of $\nabla_{T(p,6q+5)}$ are divisible
- None of $\nabla_{T(p,6q+6)}$ are divisible

Conclusion

Proofs

1. Divisibility of Knot Polynomials by casework
2. Proof of recurring cases where prime Torus knots polynomials are divisible by the trefoil polynomial

Knot Property Relationships

1. Surjective Homomorphism $\iff$ Tricolorability
2. Divisibility of Alexander Polynomial $\implies$ Tricolorability
3. Surjective Homomorphism $\implies$ Divisibility of Alexander Polynomial

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